NEEDLE: A Live Benchmark of Web Search APIs for AI Agents

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Abstract

NEEDLE measures web search APIs on five tasks: news, finance, papers, law and rare-word queries. It draws queries and reference answers from public sources on each run, except for a fixed rare-query stream, and scores the returned titles, URLs and snippets. We study 26 API configurations. Title queries hide large differences: 23 configurations achieve scholar recall@10 of .71–.98 from titles, but .08–.72 from body keywords. Blinded LLM raters broadly agree with the relevance judge. Code, data and dashboard are public; links are withheld for review.

1 Introduction

AI agents use web search APIs to find evidence for their answers. The API controls which sources the agent can inspect and what it sees before opening a page: a title, URL and snippet. End-to-end agent benchmarks combine this retrieval step with query planning, page reading and answer generation. They do not isolate the quality of the search API itself.

NEEDLE evaluates this step directly: one query in, up to k results out, scored on the evidence returned to the agent. It tests the live web rather than a fixed corpus, and draws fresh queries from public sources to track retrieval quality over time. This reduces dependence on fixed test sets that can leak into training data (Sainz et al., 2023; Balloccu et al., 2024), though registry facts and rare queries can repeat.

The benchmark combines five tasks, automatic scoring and a pooled ceiling: the best results found by any tested configuration. We report:

- Results for 26 configurations, including paired tests on 12.4k query-runs shared by the 21 main engines.
- Paired scholar query types that expose differences between title and full-text retrieval.

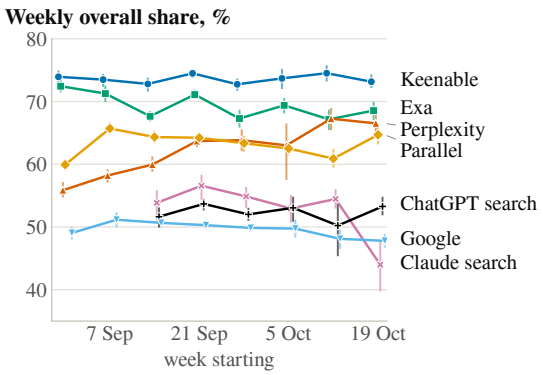


Figure 1: Overall share by ISO week, 31 August–25 October 2026. ChatGPT and Claude search start on 14 September; the pool grows from 18 to 25 configurations. Points are query-weighted means; bars are 95% run-level bootstrap intervals. Scores follow the dashboard rules.

- Checks of the judge, matchers, ceiling and failure rules, with samples and labels for readers to inspect.

2 Related Work

Live and API benchmarks. LiveBench and LiveSearchBench draw new items from dated sources but score model answers (White et al., 2025; Zhou et al., 2026a). BrowseComp scores agents end to end (Wei et al., 2025); FreshStack and BrowseComp-Plus fix the corpus to isolate retrieval (Thakur et al., 2025; Chen et al., 2026). API comparisons use static questions under one agent (Selvam and Ghosh, 2026; Artificial Analysis, 2026), while web engine audits use fixed query sets (Lewandowski, 2015; Kirsten et al., 2026).

Scorers and tasks. LLM assessors and simulated known-item queries can reproduce system rankings (Thomas et al., 2024; Upadhyay et al., 2025; Azzopardi et al., 2007; He et al., 2025). Domain benchmarks cover literature, caselaw and finance search (Ajith et al., 2024; Hou et al., 2025;

Hu et al., 2026); full text helps retrieval beyond abstracts (Lin, 2009). NEEDLE combines these methods with fresh queries and isolated API scoring. Its contribution is the live comparison and paired diagnostics across APIs, not a new scorer.

3 The Benchmark

A *configuration* is an API at one setting. We test 26 configurations from 20 vendors, including five mode variants, and call each an *engine*. A *query-run* is one query issued in one run. Appendix A gives the API settings and data sources.

3.1 Queries

Table 1 summarizes the run schedule and scoring metrics.

News. Each hour an LLM writes 20 event queries from news feeds and Google Trends, rejecting evergreen items.

Finance. Queries ask for company facts from Wikidata (Vrandečić and Krötzsch, 2014) and GLEIF, quarterly 10-Q figures from SEC XBRL, or filings by description.

Scholar. Each recent arXiv or Europe PMC (Europe PMC Consortium, 2015) paper yields four paired known-item queries (Azzopardi et al., 2007): its normalized *title*; *body keywords* for a detail absent from title and abstract; a *body question* about that detail; and a *tip-of-tongue* description (Arguello et al., 2021) without names or exact values. All four must pass the filters.

Legal. CourtListener opinions yield caption queries; eCFR sections yield queries based on verbatim spans. Filters reject citation leaks.

Rare. We sample a fixed public stream of English agent queries containing rare words (Ning et al., 2026; Li et al., 2026; Zhou et al., 2026b). Table 2 gives query examples; Appendix B gives the filters.

At least half of legal queries and finance money and filing queries carry quotes, `site:`, `after:` or `before:`. The client maps operators to native filters where available.

3.2 Scores

News and rare. GPT-6 rates each distinct result from 0 to 4 using a shared query profile, with checks for relevance, coverage, freshness and literal operator compliance. The scale adapts the Needs Met levels of Google (2025). Ratings 0–4 map to gains $g = \{0, 0, .117, .667, 1\}$, putting high weight on strong results (Kanoulas and Aslam, 2009). Snippets have a 2,000-character cap.

Task	Cadence	Queries/run	k	Metric
news	hourly	20	5	nDCG@5
finance	daily	120	5	recall@5
scholar	daily	up to 420	10	recall@10
legal	daily	40 + 0–20	5	recall@5
rare	daily	100	5	nDCG@5

Table 1: Benchmark tasks, run frequency and primary metrics. k is the number of search results requested per query. Query counts can fall below the targets after source and query filtering. Legal combines 40 CFR queries with caption queries for recent court opinions.

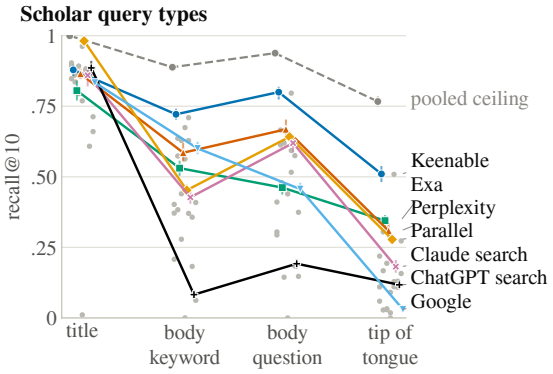


Figure 2: Scholar recall@10 by query type: 31 runs, 29 September–28 October 2026; 2,600–3,005 query-runs per line and type. Lines show seven main engines, grey dots the other 19 configurations, and dashes the pooled ceiling. Points are query-weighted means; bars are 95% run-level bootstrap intervals.

For each query, the *ceiling* keeps the best rating per normalized URL across all configurations, variants included, and sorts these ratings (Spärck Jones and van Rijsbergen, 1975; Zobel, 1998). With r_i the rating at rank i, we compute nDCG@5 (Järvelin and Kekäläinen, 2002) as

$$\mathrm{DCG}_e(q) = \sum_{i=1}^{5} \frac{g(r_i)}{\log_2(i+1)},$$

$$\mathrm{nDCG}_e(q) = \frac{\mathrm{DCG}_e(q)}{\mathrm{IDCG}(q)}.$$

IDCG is the DCG of the top five pooled ratings. Missing ranks contribute zero. Repeated normalized URLs receive zero gain except on `site:` queries.[1] We drop a query-run from all engines if any judgment fails or no result reaches rating 2.

[1]The pool still keeps each URL once, so this exception can yield nDCG@5 above 1. It affects 3 of 1,748 rare `site:` engine–query pairs in the main window.

Task	Query	Reference answer or profile
news	Paramount Warner Bros takeover closes 2026	closing date; ephemeral event
finance, fact	nvidia number of employees	42,000, ±15%
finance, money	"servicenow" q3 fiscal 2024 operating income	$418M, ±0.1%
scholar, title	optimally selling heterogeneous objects by quantity	arXiv 2610.36244
scholar, body	gap rent regularity	same paper
legal, CFR	"modify or waive any" county committees authority	7 CFR 1417.2
rare	"Krasnogvardeysky District" Russia	judged; rare word has 7 pieces

Table 2: Sample generated queries. News and rare have no reference answer; the judge profile specifies what a result must contain.

The task score is the mean over retained query-runs; the judged ceiling has mean 1.

For example, A returns ratings 4 and 2; B returns 3 and 4, with the 4-rated URL shared. The pool contains 4, 3, 2, so IDCG $= 1 + .667/\log_2 3 + .117/2 = 1.479$. A scores .73 and B .88: the denominator uses the shared pool, not each engine's own best ordering.

Scholar, legal and finance. Recall@k counts queries with a matching result in the top k. Scholar matches arXiv ids, DOIs and PMIDs; legal matches citations, dockets and identifiers in URLs and snippets. Caselaw recall measures case identity, not necessarily the opinion page. Finance checks whether the snippets contain the answer, with a 0.1% money band and a 15% headcount band. It reads table figures and parenthesized losses. An LLM may upgrade a non-money miss, but cannot revoke a hit; a failed backstop judgment excludes an engine's query only if no result matches. The ceiling is a hit by any engine. Appendix F checks matcher error.

3.3 Overall share and failures

For bench metric M_b, an engine's overall share is

$$S(e) = \frac{1}{5}\sum_{b=1}^{5} \frac{M_b(e)}{M_b(\text{ceiling})}.$$

The benches have equal weight. The ceiling is the best the tested engines found, not the best the web contains.

Table 3 uses dashboard scores from the 276 runs in which all 21 main engines took part. On finance, scholar and legal, account-side failures (our quota, rate limit, login or credit) are excluded; other failures score zero. On news and rare, every failure in this window scores zero. These exclusions remove up to 28% of an engine's query-runs.

Comparing two engines. We estimate the difference $S(a) - S(b)$, not whether their individual confidence intervals overlap. Within each task, we sample whole runs with replacement 2,000 times, drawing the same runs for every engine (Efron and Tibshirani, 1993). Each sample preserves within-run dependence and each engine's exclusions; we recompute query-weighted task and ceiling means, then overall shares. We use the bootstrap standard error to form a normal-approximation interval for each difference. Bonferroni correction accounts for testing all $\binom{21}{2} = 210$ pairs at once, at a 5% familywise level (Carterette, 2012). An interval that excludes zero supports ordering that pair under these scoring rules. The intervals measure sampling uncertainty, not scorer error. Appendix C gives the interval formula and checks other sampling units and failure policies.

3.4 Judge validation

We sample 100 news and 100 rare queries, with six distinct top-five URLs per query (1,200 results). Two Claude Opus 5.5 raters, Gemini 4 Argon and Grok 4.7 grade these with engine names hidden and a common written guideline. Claude raters also see the first 3,000 page characters. The Claude raters agree at quadratic-weighted $\kappa = .94$ (Cohen, 1968); Gemini and Grok agree with their consensus at .92 and .90. Agreement with the production judge is .77–.79 (Table 4). A separate snippet-only pass on 400 items finds that adding page text changes 25% of grades.

To test rankings on the same evidence as production, Gemini and Grok also grade 100 complete query pools (50 per task): 5,698 unique results from all 21 engines, using snippets only. Each gives engine-rank Spearman .94 against the production judge (95% intervals .89–.97 and .87–.96). All three retain the same top five in different orders; none separates Keenable from Exa on this sample. These checks measure model agreement, not human validation. Appendix E gives the audit protocol and checks sensitivity to evidence length

Engine	Overall share, %		Scholar	News	Finance	Rare	Legal	Failed, %	Latency
	mean	95% CI	R@10	nDCG@5	R@5	nDCG@5	R@5	requests	p_{50}, ms
ceiling (pooled)	*100*		*.901*	*1*	*.993*	*1*	*.971*		
Keenable	73.2	72.3–74.1	**.731**	.609	.903	**.508**	**.797**	0.0	135
Exa	68.5	67.4–69.4	.563	.599	**.922**	.476	.771	0.3	2,132
Perplexity	66.9	65.3–68.5	.578	.640	.895	.468	.674	0.0	433
Parallel	64.9	63.6–66.1	.587	.535	.907	.387	.734	0.8	2,660
Brave (LLM ctx)	64.6	63.4–65.6	.533	**.642**	.858	.454	.658	0.0	1,112
Linkup	57.6	56.1–58.9	.470	.567	.797	.400	.570	0.0	1,212
Brave	57.3	56.2–58.4	.488	.557	.769	.354	.621	0.0	487
Context	57.1	55.4–58.6	.598	.469	.782	.339	.576	0.0	866
DeepSeek	56.9	55.8–57.9	.271	.635	.818	.387	.676	0.0	551
You.com	55.8	54.7–56.9	.471	.553	.747	.343	.602	0.0	486
ChatGPT search†§	51.5	49.4–53.1	.300	.340	.809	.366	.697	11.1	43,831
Kagi§	51.3	49.9–52.6	.560	.301	.728	.275	.617	7.6	1,275
Firecrawl	50.7	49.2–52.0	.488	.286	.820	.265	.598	0.0	688
Octen	50.2	48.8–51.7	.307	.511	.730	.322	.586	0.0	197
Google (Serper)	47.7	46.8–48.6	.475	.262	.738	.255	.582	0.0	1,785
Claude search†§	45.1	42.7–47.6	.498	.276	.685	.195	.525	35.9	33,906
TinyFish§	44.5	43.0–45.9	.429	.256	.730	.220	.521	29.0	1,904
Bing (SearchAPI)	44.1	41.1–46.8	.416	.288	.675	.173	.586	0.4	2,008
Jina	43.8	42.4–45.1	.492	.210	.726	.163	.523	4.9	17,272
Tavily	42.6	41.5–43.7	.303	.303	.828	.284	.361	0.1	1,999
Ceramic	14.7	13.9–15.7	.001	.040	.441	.109	.137	0.1	99

Table 3: Results on 276 shared runs, 17–28 October 2026 (12.4k query-runs, 21 engines); the ceiling also includes variants. Bench cells are absolute metrics, best in bold; italics mark the ceiling. Failed counts all request failures. Latency is the per-run median on news. †Server-side search tool; latency includes answer generation. §Account-side exclusions leave rank uncertain (Appendix C).

Quadratic-weighted κ	News	Rare	All
Claude A vs. B	.94	.94	.94
Judge vs. Claude consensus	.72	.80	.77
Gemini vs. consensus	.91	.92	.92
Grok vs. consensus	.90	.88	.90
Gemini vs. judge	.74	.81	.78
Grok vs. judge	.75	.81	.79

Table 4: Agreement on 1,200 blinded results, 600 per bench. Consensus is the rounded mean of the two Claude page grades. The judge–consensus 95% query-bootstrap interval is .74–.81.

and literal quotes.

4 Results

The data support most pairwise rankings, not every rank. The adjusted score-difference intervals exclude zero for 178 of 210 engine pairs. We cannot reliably order the remaining 32 pairs from this sample; this does not establish that their scores are equivalent. The overall lead also depends on the task mix: scholar contributes 3.7 points of Keenable's 4.7-point lead over Exa. Without scholar, the lead is 1.2 points (Appendix C).

Body queries expose gaps that title queries hide. Over 31 scholar runs, 23 of 26 configurations reach recall@10 .71–.98 from titles. The same 23 span .08–.72 on body keywords (Figure 2; Table 7). Keenable leads at .72, followed by its realtime mode (.71) and Kagi (.67). Appendix D gives paired intervals on common papers.

Weekly ranks are stable, not fixed. Adjacent weekly rankings agree at Kendall's τ_b (Kendall, 1945) .81–.94. Keenable leads all eight weeks, but lower ranks change (Figure 1). Appendix C checks changes in the engine pool, account failures and dispatch times.

Limitations

Conflict of interest. The authors' organization builds an evaluated engine. The code, prompts, clients and per-query outputs are public so readers can audit the claims. All engines receive the same queries, snippet cap, request pacing and judge; each keeps one setting throughout the main window. We cannot tell whether a vendor changed its index for the benchmark or for other reasons.

Evidence and judge. Scores measure the API response, not the fetched page. Short snippets can hide relevant content (Turpin et al., 2009; Kazai et al., 2022), and identifier matching can miss a target without its id. On 400 sampled results, page text changes 25% of grades. News and rare depend on one judge; all audit raters are LLMs, and none provides human validation, which LLM labels cannot replace (Faggioli et al., 2023; Clarke and Dietz, 2025). Shared rules can cause shared errors. Query-

term stuffing can fool judges (Alaofi et al., 2024); vendor and model biases can affect scores (Balog et al., 2025; Soboroff, 2025). The judge shares a vendor with ChatGPT search, and some audit raters share one with Claude search; LLM evaluators can favor their own outputs (Panickssery et al., 2024).

Matcher error. Finance table figures can match the wrong period when snippets omit column headers. A fresh audit confirms only 70% of table-reading hits; most others are unresolved, not contradicted. The pooled error estimates cannot show each engine's error rate. Caselaw recall credits pages about the case, not just the opinion. Appendix F gives the finance checks.

Coverage and weights. Queries are English and US-centric. Equal bench weights do not estimate agent traffic, and the lead depends strongly on scholar. Default API tiers need not be each vendor's best. ChatGPT and Claude search return sources through answer-generating tools, so their quality scores are comparable but their latency and cost are not. The ceiling depends on the engine pool (Buckley et al., 2007); compare shares within a common window.

Failures and timing. Excluding account-side failures assumes those requests would score like the rest. Without that assumption, several middle ranks remain uncertain. News and rare also count our credit lapses as zeros in the main window. Engines process queries at different speeds, so slow engines see news later. A five-minute dispatch filter leaves the news order unchanged. Vendor redirect links can break URL matching.

Repetition and reproducibility. Registry facts and the fixed rare stream can repeat. Public sources and queries permit tuning or special handling of benchmark traffic. The studied runs archive queries, reference answers, results and judgments, but lack API publication dates and each run's code revision. Stored ratings reproduce the tables; re-judging cannot exactly reproduce the inputs. Appendix A gives the source terms.

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A API Settings and Data Sources

All engines receive the same query and requested result count. Each engine processes one request at a time; engines run concurrently. Requests have a 30-second timeout and up to three attempts, with exponential back-off from 1 to 30 seconds on HTTP 429, 5xx and timeouts. Short result lists score as returned. Snippets have a 2,000-character cap.

The generators and judge use GPT-6 Luna (`openai/gpt-6-luna`) through OpenRouter, temperature 0 and high reasoning effort. Table 5 lists the search clients and settings. Supporting scripts, samples and labels are under `paper/analysis/`; the manifest identifies each analysis's inputs.

The rare stream draws from DeepResearchGym (Coelho et al., 2026; Ning et al., 2026) (59%, CC BY 4.0), OpenResearcher (Li et al., 2026) (39%, MIT) and LRAT-Train (Zhou et al., 2026b) (1%, no stated license). Source terms govern query texts; our code is MIT and our CC BY 4.0 covers results, judgments and metadata, not source text. LRAT contributes 17 of 1,200 main-window rare queries; removing them changes no mean rare nDCG@5

Engine	Endpoint	Variant or non-default parameters
Keenable	api.keenable.ai	default; realtime variant
Exa	api.exa.ai	`type=auto`; instant and lite variants
Google	google.serper.dev (Serper)	default
Bing	searchapi.io (SearchAPI)	engine `bing`
Brave	api.search.brave.com	web search; LLM-context endpoint
Parallel	api.parallel.ai	default processor; turbo variant
Perplexity	api.perplexity.ai	search API; fast variant
Tavily	api.tavily.com	default
Linkup	api.linkup.so	default
TinyFish	api.search.tinyfish.ai	default
Octen	api.octen.ai	default
Ceramic	api.ceramic.ai	default
You.com	ydc-index.io	default
Firecrawl	api.firecrawl.dev	search endpoint, default
Kagi	kagi.com/api/v1	default
Jina	s.jina.ai	default
Context	api.context.dev/v1	default
DeepSeek	api.deepseek.com	`/v1/tools/search`, default
Claude search	api.anthropic.com	`claude-sonnet-5`, `web_search`
ChatGPT search	api.openai.com	`gpt-5.5`, `web_search`, `search_context_size=low`

Table 5: API configurations. Clients live in `src/needle/shared/search/`; clients also implement the native query filters.

by more than .003. Queries can contain personal information. Release contacts accept private removal requests; API output records remain subject to vendor terms.

B Query Filters

Scholar samples five domains (computer, physical, life, social and health sciences) and three age bands (7 days, 30 days, 1 year), targeting seven accepted papers per cell. Each paper must pass all four filters:

- **Title:** at least four words, plus a digit, acronym, hyphenated term or five content tokens.
- **Body keywords:** at least three words and two content tokens absent from title and abstract.
- **Body question:** at least eight words and the same novelty test as body keywords.
- **Tip-of-tongue:** at least twelve words; no quotes, decimal, five-digit number or coined title/abstract term; at most two content tokens shared with the title.

An instrumented run accepts 323 of 572 candidate papers (56%). The effect of rejecting other papers on engine order is untested.

News draws from 124 feeds and Google Trends RSS for 52 US geographies; prompts reject evergreen items and request two-to-six-word event queries. Legal captions come from 14 federal courts on CourtListener. CFR queries contain one verbatim two-to-eight-word span but no citation or target section number. Only the quoted variant retains quotation marks. Rare queries must pass fastText English language ID (Joulin et al., 2017, 2016) and contain a word split into at least five BERT WordPieces (Wu et al., 2016; Devlin et al., 2019).

C Statistical Checks and Ranking Sensitivity

Paired intervals. The main window has 12 scholar, 232 news, 10 finance, 12 rare and 10 legal runs. For pair (a, b), let $\widehat{\Delta}_{ab} = S(a) - S(b)$ and let s_{ab} be the standard deviation of its 2,000 bootstrap differences. The adjusted interval is

$$\widehat{\Delta}_{ab} \pm 3.683\, s_{ab}.$$

This is a normal approximation, not a percentile bootstrap interval. The implemented critical value is slightly more conservative than 3.675, the two-sided Bonferroni value for 210 comparisons at familywise level .05. Individual engine intervals in Table 3 instead use the 2.5th and 97.5th bootstrap percentiles. Their overlap is not a test of the difference between two engines.

Dependence and missing requests. We repeat the comparisons by resampling days, and by clustering exact query text (target paper for scholar) across runs. The latter checks repeated targets, which day resampling alone does not address. We also restrict the data to query-runs with no account-side failure in any main engine. Of the 178 pairs

Engine	Historical	Consistent policy	Matched	Δ matched	95% CI for Δ
Keenable	73.2	73.2	74.7	+1.6	+0.4, +2.7
Exa	68.5	68.6	67.5	−1.1	−2.3, +0.2
Perplexity	66.9	66.9	66.2	−0.7	−2.5, +1.1
Parallel	64.9	65.1	63.0	−2.1	−3.5, −0.7
Brave (LLM ctx)	64.6	64.6	67.1	+2.5	+1.1, +3.9
Linkup	57.6	57.6	59.3	+1.7	−0.1, +3.8
Brave	57.3	57.3	59.1	+1.8	+0.2, +3.2
Context	57.1	57.1	56.2	−0.9	−3.0, +1.1
DeepSeek	56.9	56.8	58.0	+1.1	−0.6, +2.9
You.com	55.8	55.8	57.4	+1.6	+0.3, +2.9
ChatGPT search	51.5	52.4	51.9	−0.5	−2.0, +0.8
Kagi	51.3	52.1	52.3	+0.1	−0.8, +1.2
Firecrawl	50.7	50.7	50.1	−0.5	−1.9, +0.9
Octen	50.2	50.2	49.1	−1.1	−2.7, +0.6
Google (Serper)	47.7	47.7	47.4	−0.3	−1.5, +0.9
Claude search	45.1	51.4	52.6	+1.2	−0.4, +2.9
TinyFish	44.5	45.1	44.7	−0.4	−2.0, +1.2
Bing (SearchAPI)	44.1	44.1	48.3	+4.1	+1.5, +7.5
Jina	43.8	43.8	43.2	−0.5	−2.2, +1.2
Tavily	42.6	42.6	42.5	−0.1	−1.2, +1.3
Ceramic	14.7	14.7	14.7	−0.1	−1.1, +1.2

Table 6: Overall shares in percent. Historical repeats Table 3. Consistent policy excludes account-side failures on every task and scores other failures zero, using 12,313 available query-runs. Matched restricts these to 4,048 query-runs with no account failure in any main engine. Δ is Matched minus Consistent policy, in percentage points, with a paired run-bootstrap interval.

whose primary adjusted intervals exclude zero, 164 also exclude zero under both day resampling and this matched-support check.

Table 6 separates the failure-policy change from the change in query support. Per-query files cover 274 of the 276 runs; matching retains 4,048 of 12,313 query-runs. With one account-failure policy, Claude search rises from 45.1 to 51.4 overall; matching changes it to 52.6. The matched order has $\tau_b = .89$ against the main table, with Keenable and Exa still first and second. Its difference intervals use 1,000 paired run-bootstrap resamples. If excluded requests instead score zero or at the ceiling, ranks 1–10 stay fixed under both extremes, but Claude search and TinyFish can occupy ranks 11–20. Missing requests therefore limit some rank claims.

Normalization, weights and timing. Removing an engine from the pool changes its overall share by at most 0.8 points. Dropping any one task preserves the leader but changes the Keenable–Exa gap from 1.2 to 6.4 points. The lead's size is sensitive to task choice even when the leader stays the same.

Among the 15 configurations present in all eight plotted weeks, adjacent rankings have $\tau_b = .85$–$.94$. Fixing the news pool to those 15 changes no sampled weekly order. Restricting news comparisons to requests sent within five minutes of the first engine retains 96% of scored pairs, leaves the order unchanged and moves no mean nDCG@5 by more than .010. These checks address changing engine participation and dispatch lag.

D Scholar Query Types and Paired Differences

Table 7 reports every configuration and its sample size, rather than only the seven lines in Figure 2. In the main window, Keenable's recall lead over Exa is .38 for 7-day papers, .10 for 30-day papers and .03 for one-year papers; it is negative in health sciences ($-.10$). The advantage is not uniform across age or domain.

For paired query-type comparisons, we retain 2,629 paper-runs from 29 runs where Keenable, its realtime mode, Kagi, Exa and ChatGPT search all have scores for all four types (10,516 query-runs). Account-side failures exclude the paper from all five; other failures score zero. We use stored identifier hits and 2,000 paired run-bootstrap resamples.

Body-keyword minus title recall is $-.154$ [$-.178$, $-.132$] for Keenable and $-.802$ [$-.833$, $-.772$] for ChatGPT search. On body keywords, Keenable's paired lead is .012 [.007, .018] over realtime, .048 [.021, .074] over Kagi, .199 [.174, .226] over Exa and .644 [.619, .669] over ChatGPT search. These pointwise 95% intervals use

Engine	Title	Body keyword	Body question	Tip of tongue	Runs	Query-runs
ceiling (pooled)	*1*	*.888*	*.938*	*.767*	*31*	*12,020*
Parallel	**.981**	.453	.642	.278	31	12,020
Kagi	.962	.674	.640	.091	30	11,599
Bing (SearchAPI)	.903	.402	.411	.060	31	12,020
Parallel Turbo	.894	.208	.294	.110	31	12,020
Brave (LLM ctx)	.893	.440	.620	.194	31	12,020
ChatGPT search	.886	.083	.192	.118	30	11,016
You.com	.883	.370	.593	.129	31	12,020
Brave	.881	.383	.613	.167	31	12,020
Keenable	.879	**.721**	**.800**	**.510**	31	12,020
Linkup	.878	.357	.575	.130	21	7,898
Keenable realtime	.877	.709	.796	.509	31	12,020
Perplexity	.865	.585	.668	.309	30	11,636
Jina	.862	.629	.470	.030	31	11,716
Perplexity Fast	.860	.574	.666	.305	30	11,636
Claude search	.860	.427	.620	.182	27	10,400
Context	.856	.628	.516	.111	31	12,020
Exa Lite	.848	.371	.305	.219	1	420
Firecrawl	.836	.636	.472	.028	31	12,020
Google (Serper)	.834	.601	.457	.030	31	12,020
Exa	.805	.531	.462	.345	31	12,020
Tavily	.795	.279	.144	.033	31	12,020
TinyFish	.769	.563	.440	.018	31	3,472
Exa Instant	.713	.409	.374	.273	31	12,020
Octen	.660	.063	.147	.157	31	12,020
DeepSeek	.608	.183	.184	.109	12	4,820
Ceramic	.011	.000	.000	.000	31	12,007

Table 7: Scholar recall@10 by query type, 31 runs, 29 September–28 October 2026. The ceiling has 3,005 query-runs per type; the last column sums all four types. The 31 runs contain 332–420 queries each, totaling 12,020 rather than 31×420: 420 is a target, not a fixed sample size. Run counts show scored support; Exa Lite has only one run. Counts also fall after account failures. Bold marks the best configuration per type (`bucket_table.py`).

common papers, unlike the available-case means in Table 7; they do not measure identifier-matcher error. Full contrasts and replay inputs are in `audit/scholar_paired.json` and its companion JSONL file.

E Judge Sensitivity Checks

Audit protocol. The 200 queries in Section 3.4 come from 17–29 October runs with at least 19 engines. Each Claude rater uses a separate random order in 24 sessions of 50 items; Gemini and Grok grade one item per call at temperature 0. All use the same written guideline. Raters assign mean grades of 1.87–1.97 versus the production judge's 1.60, so agreement does not imply identical grade calibration.

Evidence length. The staged snippet-versus-page check changes 25% of 400 grades (95% interval 21–29%), but the mean grade change is only +.05 [−.01, +.11]. Positive and negative changes can cancel in the mean.

On 803 news results from eight queries, a Claude Opus 5.5 judge grades both full snippets and versions capped at 200 characters. The resulting engine orders agree at $\tau_b = .73$ (95% interval .40–.84); mean absolute share change is .084. Exa and Keenable lose, while Bing and Google gain. The sample does not distinguish the top two at either length. Snippet length is thus part of the measured API quality, not a controlled property of the underlying page.

Literal quotes. Of 1,200 rare query-runs, 504 contain quotes. On unquoted queries, Exa, Perplexity and Brave (LLM ctx) lead; Keenable is fourth. Replacing the rare score with its unquoted-only value gives overall $\tau_b = .96$ and a 3.3-point Keenable lead over Exa. Literal phrase matching therefore affects the rare ranking even when the overall order changes little. These checks use LLM judgments, not human labels.

F Finance Matching and Checks

Matching rules. Money must fall within 0.1% of the reference value; headcounts within 15%. A scale word fixes a money figure's scale. Bare figures, including parenthesized losses, can mean thousands or millions only when the preceding 200 characters contain a currency or financial cue.

Years, percentages and multiples do not qualify. Diluted EPS also needs "EPS", "per share" or "earnings per". Money gets no LLM backstop; other fields can receive an upgrade, never a revoked hit. The full rules are in `src/needle/finance/canon.py`.

Why the narrow money band? An LLM audit of 399 results checks company, period and line item. Money-hit precision without table reading is .93 at the 0.1% band and .88 at 2%: the wider band admits more wrong periods and related line items. Money backstop upgrades have precision .36, which motivates excluding them. These are model judgments, not human gold labels.

Unseen-query check. A separate sample has 307 unseen queries (146 money, 161 other), tested on 17 configurations. An LLM checks the first matching result for 335 table-reading hits, 100 exact-band money hits and 50 other-field hits, using at least the full 2,000-character input. Exact-band precision is .93. Of 335 table-reading hits, 235 confirm the correct figure and period (.70; 95% Wilson interval .65–.75; Wilson, 1927), 95 omit period labels and five contradict the answer. Counting unresolved hits as wrong gives precision .70; counting only contradictions gives .99. Confirmation rates differ by engine: for example, 13/43 for Firecrawl and 42/44 for Exa. Pooled precision cannot establish each engine's error rate.

Effect on reported ranks. Discounting each engine's main-window table-reading hits by its fresh unconfirmed fraction lowers finance recall by up to 6.0 points; overall ranks 1–10 stay fixed. Dropping table reading entirely lowers finance recall by 0.1–7.5 points, with finance $\tau_b = .84$ and overall .97; overall ranks 1–3 stay fixed. This bounds one scoring choice, not all matcher error. The fresh sample has more money queries than production and lacks the non-money backstop, so it cannot replace the main table.